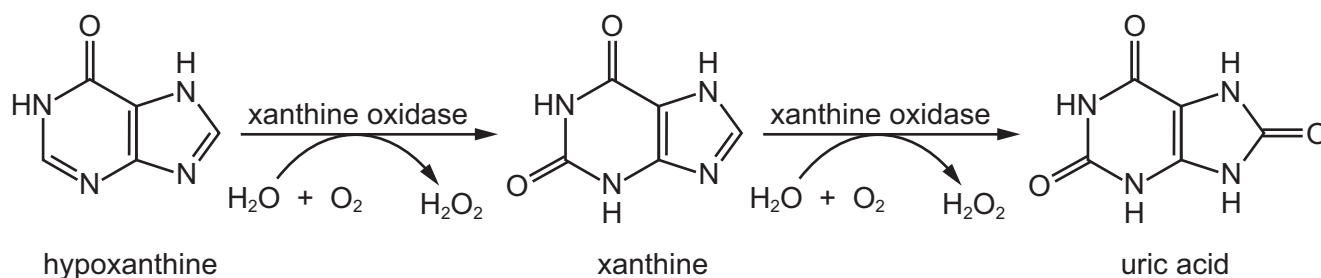
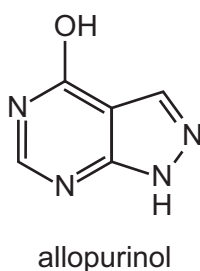


- 13** Gout is a type of arthritis in which small uric acid crystals form inside and around the joints. It causes sudden attacks of severe pain and swelling.

The diagram shows how uric acid is formed from hypoxanthine catalysed by the enzyme xanthine oxidase.



Gout can be treated using a drug called allopurinol which has a similar shape to hypoxanthine.



What can be concluded from this information about how allopurinol prevents the formation of uric acid?

- A** It binds to the active site of xanthine oxidase instead of hypoxanthine, resulting in reduced production of uric acid.
- B** It binds to another part of xanthine oxidase and this changes the shape of the active site.
- C** It disrupts the hydrogen bonds within xanthine oxidase so it denatures and the active site is no longer complementary to hypoxanthine and xanthine.
- D** It hydrolyses the peptide bonds within xanthine oxidase to change the shape of the active site.